Two forbidden patterns bound the boundary intersections of two simple heptagons

A Lean-verified bound $|\partial P \cap \partial Q| \leq 38$ for simple heptagons that share no vertex, and the vertex-coincidence reduction needed for $f(7,7) = 38$

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Abstract

Let $f(m,n)$ be the maximum number of points at which the boundaries of a simple m-gon and a simple n-gon can meet, over pairs whose boundaries meet in finitely many points. Dillencourt, Mount and Saalfeld posed the problem in 1993 and settled the cases where m or n is even; $m = n = 7$ is the smallest open odd–odd case. We prove that if two simple heptagons *share no vertex* and their boundaries meet in finitely many points, then the boundaries meet in at most 38 points, and that 38 is attained. A vertex of one heptagon may lie in the interior of a side of the other, and no collinearity condition is imposed; in particular the bound covers every pair in general position. This theorem is verified by the Lean 4 kernel. Deducing $f(7,7) = 38$ for all pairs, including pairs that share vertices, requires in addition a reduction that removes vertex–vertex coincidences without losing intersection points or simplicity. Ackerman, Keszegh and Rote (§9.1 of the journal version) give such a reduction. We use it but do not formalize it, and we have not completed a check that its conclusion — in particular that the perturbed polygons remain simple — holds for the exact definitions used here (§8). The equation $f(7,7) = 38$ is therefore stated here as a consequence of that external reduction, with this qualification.

The mathematical content is a reduction. Rather than enumerate the 13,003,620 candidate crossing patterns with 40 intersections — the number of 9-subsets of the 7×7 grid meeting every row and column, or about 33,000 after quotienting by symmetry — we exhibit *two* local patterns, prescribing 8 and 14 side-pair incidences respectively (the second together with seven self-non-crossing conditions), and show that every candidate pattern with 40 intersections and every one with 42 contains one of them. Each of the two is non-realizable, and a degree argument over 28 cases, itself discharged by machine, reduces the whole exclusion to those two facts. That discharge is not hand-checkable (Remark 4.2).

The mathematical assertions of §§2–7, the vertex–side reduction of §8.1, and Theorem 1.2 itself are verified by the Lean 4 kernel from primitive definitions, with no published result about $f(m,n)$ used as an input. Two negative observations in §8.1 are analytic and are *not* machine-checked. We distinguish throughout (a) what is kernel-checked, (b) what follows from it together with the external vertex-coincidence reduction, and (c) what remains unresolved in this project — including whether the formal definitions express the intended objects, which is a matter for a human reader.

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*This work was carried out with substantial generative-AI assistance; see the disclosure that precedes the appendix.

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1 Introduction

1.1 The problem

Write $f(m, n)$ for the maximum of $|\partial P \cap \partial Q|$ over *all* pairs of simple polygons with m and n sides whose boundaries meet in finitely many points. The finiteness restriction cannot be dropped

— two collinear overlapping sides share infinitely many points — but no further restriction is imposed.

We will also need a technical condition. Two simple polygons are in *general position* if no vertex of either lies on the boundary of the other and no three of the combined $m + n$ vertices are collinear. This is *not* part of the definition of f used here, and the distinction matters. Several sources state the problem under general-position assumptions: Černý et al. [5, §2] assume that all segment endpoints are in general position, and [2, §3] assume finitely many intersection points, then assume that no vertex of one polygon lies on the other and justify that assumption in their §9. Theorem 1.2 assumes neither half of general position: it assumes only finiteness and that the two polygons share no vertex. It therefore determines the maximum over general-position pairs outright. It does not by itself determine $f(7, 7)$ as defined above, because f also ranges over pairs that share vertices; that step is Proposition 8.4, which we take from the literature and do not prove.

Problem (Dillencourt–Mount–Saalfeld [6]). Determine $f(m, n)$.

They determined $f(m, n)$ when at least one of m, n is even. The odd–odd cases are open. A 2012 preprint of Günther [7] claiming $f(n, n)$ for odd n was withdrawn by its author, the stated reason being "an unrecoverable error in the proof of Lemma 2", with a counterexample found to a weaker form of that lemma. The smallest open case is $m = n = 7$.

Remark 1.1 (A caution for anyone searching the literature). Several web search engines, asked about this problem, return summaries asserting that the odd–odd conjecture *is* proved. Those summaries paraphrase the *withdrawn* preprint. We mention it because we were shown such a summary during preparation.

Two elementary facts frame it. Each side of P meets each side of Q at most once in general position, so $I \leq 49$; and since ∂P and ∂Q are closed curves meeting transversally, I is even. For pairs in general position, Černý et al. [5] give $I \leq 40$, and 38 is attained. So, in general position, the question is whether 40 is attainable.

1.2 What is proved here

Theorem 1.2 (Main)**.** *Let P and Q be heptagons in the plane, each simple as a point set (Appendix A), whose boundaries meet in finitely many points, and such that no vertex of P is a vertex of Q. Then*

$$|\partial P \cap \partial Q| \;\leq\; 38,$$

and 38 is attained by such a pair. No condition on collinearity of vertices is assumed, and a vertex of one polygon is permitted to lie in the interior of a side of the other.

Theorem 1.2 is an unconditional theorem under its stated hypotheses, and it is kernel-checked: it is the pair of Lean declarations `Pair9Shift.points9_le_38_of_coincFree` and `Pair9Shift.isGreatest_38_of_coincFree` (§9). The finiteness of $\partial P \cap \partial Q$ is an explicit hypothesis. It cannot be dropped: two collinear overlapping sides meet in infinitely many points.

1.3 Scope

Three problems should be kept apart.

1. **Pairs in general position.** Such a pair shares no vertex, and its boundary intersection is finite (an overlap of two sides would put three vertices on a line). So Theorem 1.2 applies, and the maximum over general-position pairs is 38. This case is also kernel-checked separately, as `RealUpperBound.isGreatest_38_real`, where general position is part of the structure.

2. **Pairs with finite intersection and no shared vertex.** This is Theorem 1.2. It allows a vertex of one polygon in the interior of a side of the other, tangencies, and any collinearities.

3. **All pairs with finite intersection**, shared vertices included. This is $f(7,7)$. The condition "P and Q share no vertex" is a hypothesis of Theorem 1.2. It is a different thing from the *reduction* (Proposition 8.4) asserting that every pair can be replaced by one that shares no vertex, is again simple, again has finite intersection, and has no fewer intersection points. With that reduction, Theorem 1.2 gives $f(7,7) = 38$.

For the third problem we separate (a) what is kernel-checked: Theorem 1.2, including attainment; (b) what follows from (a) together with an external theorem: $f(7,7) = 38$, using [2, §9.1]; and (c) what this project has not resolved: whether §9.1, read for the exact definitions used here, delivers a pair that is again simple in the sense of Appendix A and again has finite intersection, in addition to the intersection-count inequality that §9.1 states. §8 sets this out. Using a published theorem does not make an argument incomplete; the qualification here concerns only that applicability point, which we have not settled.

Earlier versions of this paper assumed the stronger hypothesis (E9): no vertex of either polygon lies anywhere on the boundary of the other. That is the full conclusion of [2, §9], which splits into §9.1 (vertex–vertex coincidences) and §9.2 (vertex–side incidences). The (E9) version remains kernel-checked (`AKR9.isGreatest_38_of_E9`); under (E9) finiteness is a consequence rather than a hypothesis. It is superseded by Theorem 1.2, because the vertex–side half is proved in this paper (§8.1).

1.4 What is new

The contribution is a reduction, not a technique. Both ingredients we use — encoding realizability of a crossing pattern as an oriented-matroid satisfiability problem, and refuting it mechanically — are standard; see §10. What appears to be new is:

- **Two patterns suffice** (Theorem 4.1). A search that was 13,003,620 cases becomes two non-realizability facts plus a degree argument over 28 cases.
- **The smaller pattern is human-readable** and needs no simplicity hypothesis: it constrains three consecutive sides of each polygon and prescribes eight incidences (§3).
- **The reduction to the plane is carried out in full.** The combinatorial argument is about orientation signs; getting from there to genuine point sets in $\mathbb{R}^2$ — including removing the collinearity assumption that [2, §3] make "for convenience" and state can "be achieved easily", without giving the details (quoted in full in §8) — and removing vertices from the interiors of sides is proved and kernel-checked (§§7, 8.1).

A bounded prior-art search found no previous statement of the covering result. We stress in §10 that this is weak evidence and not a novelty claim.

2 Preliminaries

2.1 Configurations and the disjointness graph

Let P have vertices $p_0, \dots, p_6$ and Q have $q_0, \dots, q_6$, indices mod 7, with sides $P_i = [p_i, p_{i+1}]$ and $Q_j = [q_j, q_{j+1}]$. For points a, b, c write

$$\mathrm{or}(a,b,c) = (b_1 - a_1)(c_2 - a_2) - (b_2 - a_2)(c_1 - a_1),$$

twice the signed area. Sides $[a,b]$ and $[c,d]$ *cross* when

$$\mathrm{or}(a,b,c)\cdot\mathrm{or}(a,b,d)<0 \quad\text{and}\quad \mathrm{or}(c,d,a)\cdot\mathrm{or}(c,d,b)<0.$$

This is a conjunction of *strict* inequalities, a fact used repeatedly: it makes crossing an open condition.

Definition 2.1 (after [2])**.** The *disjointness graph* G^D, whose name and notation we take from [2], is the bipartite graph on $\{P_0,\dots,P_6\}\sqcup\{Q_0,\dots,Q_6\}$ with an edge P_iQ_j when P_i and Q_j do *not* cross.

In general position each side pair contributes at most one intersection point, and exactly one when it crosses, so writing I for the number of intersection points and $|E^D|$ for the number of edges of G^D,

$$I+|E^D|=49. \tag{1}$$

2.2 Two unconditional bounds

Lemma 2.2. *I is even.*

Proof sketch. ∂P and ∂Q are closed curves meeting transversally, so their mod-2 intersection number is 0: each side of P contributes to I once per transversal crossing, and summing the crossing parities around the closed walk ∂P returns to the starting value. Hence $I\equiv 0 \pmod 2$. The formal argument is carried out on the orientation data, so the Jordan curve theorem is not used — neither here nor anywhere else in this paper. □

Lemma 2.3. *$\delta(G^D)\geq 1$, and consequently $I\leq 42$.*

Proof. Fix a side P_i and let ℓ be the line through it. Since ∂Q is a closed curve, ℓ meets it an even number of times, so ℓ cannot cross all seven sides of Q; in particular P_i fails to cross at least one Q_j. The same holds with the roles exchanged, so every one of the fourteen vertices of G^D has degree at least 1 and $|E^D|\geq 7$. With (1) this gives $I\leq 42$. □

Remark 2.4 (Where simplicity is and is not needed)*.* Lemma 2.3 does *not* need simplicity, and it is worth being precise about why, because the neighbouring corollaries do.

The argument above uses only that ∂Q is a *closed* walk, which is true whether or not it crosses itself. So $\delta(G^D)\geq 1$ and $I\leq 42$ hold for any closed polygons in general position.

What does need simplicity is the *exclusion* of 40 and 42, because Theorem 3.7's seven self-non-crossing conditions are supplied by it. Here is a witness, and note carefully what it does and does not show: the 14 points

$$P:(197,86),(182,258),(152,128),(269,227),(90,125),(278,210),(132,227),$$
$$Q:(149,267),(228,177),(105,155),(278,223),(38,63),(253,284),(184,98)$$

have all 364 orientation determinants nonzero and satisfy $I=42$; both heptagons are self-crossing, and G^D is a perfect matching. This does *not* contradict Lemma 2.3 — it *attains* the bound. What it contradicts is Corollary 5.1 read without a simplicity hypothesis, which is why that hypothesis appears there and in Corollary 5.2. Lemma 2.2, like Lemma 2.3, needs no simplicity at all.

Remark 2.5. Černý et al. [5, Lemma 1] obtain $\delta(G^D)\geq 1$ from the fact that a line meets a simple 7-gon in at most 6 sides. We do not import that: the minimum-degree statement is derived directly and is machine-checked unconditionally, so their Lemma 1 is corroboration here, not a dependency.

By Lemmas 2.2 and 2.3, $I\in\{\dots,38,40,42\}$, and since 38 is attained (§6) the theorem reduces to excluding 40 and 42.

2.3 Orientation data

All of §§3–5 concerns only the signs of the $\binom{14}{3} = 364$ determinants $\mathrm{or}(v_a, v_b, v_c)$ on the combined vertex set. A *configuration* is a 14-tuple of points together with the requirement that all 364 signs be nonzero; a *crossing pattern* is the resulting 7×7 Boolean matrix of side-pair crossings. Realizability questions about patterns are therefore questions about which sign vectors arise from actual point sets — that is, about chirotopes.

3 Two forbidden patterns

The plan for the next two sections: we exhibit two local crossing patterns and show in §3.4 that neither is realizable; then §4 shows that every crossing pattern with $I \in \{40, 42\}$ contains one of them. The two patterns were *found* by computer search — minimising unsatisfiable cores — but what they are is small enough to state in a table, and Π_8 is small enough to check by hand.

3.1 The small pattern

Definition 3.1. Pattern Π_8 constrains sides P_0, P_1, P_2 and Q_0, Q_1, Q_2:

	Q_0	Q_1	Q_2
P_0	$\times$	$\times$	$\times$
P_1	$\times$	$\cdot$	$\times$
P_2	$\times$	$\times$	$*$

where $\times$ means *crosses*, $\cdot$ means *does not cross*, and $*$ is unconstrained. That is eight prescribed incidences: seven crossings and one non-crossing.

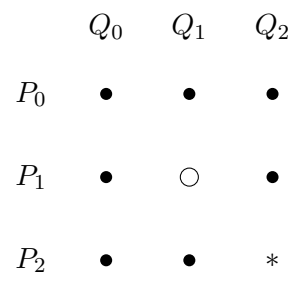


Figure 1: The pattern Π_8. A filled dot means the two sides cross; the open circle at (P_1, Q_1) means they do not; $*$ is unconstrained. Eight prescribed incidences on three consecutive sides of each polygon. No simplicity hypothesis is needed.

Theorem 3.2. *No configuration realizes* Π_8.

Theorem 3.2 assumes nothing beyond the eight incidences and the nondegeneracy of the 364 determinants: in particular it needs *no* hypothesis that P or Q is simple. It is the single most compact *statement* in this work.

An earlier version of this paper said that we offered no human-surveyable proof of Theorem 3.2, and invited a specialist to find one. That is no longer accurate for Π_8, and we give the argument here. It rests on a single general observation about two segments sharing an endpoint.

Lemma 3.3 (Shared endpoint)**.** *Let A, B, C, D, E be points with no three collinear. If the segments $[B, A]$ and $[B, C]$, which share the endpoint B, both properly cross $[D, E]$, then*

$$\mathrm{or}(A, B, D) \cdot \mathrm{or}(B, C, D) < 0 \qquad \textit{and} \qquad \mathrm{or}(A, B, E) \cdot \mathrm{or}(B, C, E) < 0.$$

Proof. Let T be the unique affine map with $TB = (0,0)$, $TA = (1,0)$, $TC = (0,1)$; it multiplies every orientation determinant by the same nonzero constant, so both displayed inequalities are unchanged by applying it, and we may assume B, A, C are those three points. Then for every $X = (X_1, X_2)$ we have $\mathrm{or}(A, B, X) = -X_2$ and $\mathrm{or}(B, C, X) = -X_1$, so the claim is exactly that D and E each have coordinates of opposite signs.

Since $[B, A]$ is the unit segment on the first axis and properly crosses $[D, E]$, the line through D and E meets that axis at a point $(t, 0)$ with $t \in (0, 1)$, and that point lies strictly between D and E. Likewise $[B, C]$ gives a point $(0, s)$, $s \in (0, 1)$, strictly between D and E. Parametrise the line as $\lambda \mapsto (1-\lambda)(0, s) + \lambda(t, 0)$, so that $\lambda = 0$ and $\lambda = 1$ are those two interior points. As both are interior to $[D, E]$, one of D, E has $\lambda < 0$ and the other $\lambda > 1$. The point at parameter λ is $(\lambda t,\ (1-\lambda)s)$ with $s, t > 0$; for $\lambda > 1$ its coordinates are $(+, -)$ and for $\lambda < 0$ they are $(-, +)$. □

Proof of Theorem 3.2. Suppose some configuration realizes Π_8. Write

$$v_k = \mathrm{or}(q_0, q_1, p_k), \qquad w_k = \mathrm{or}(q_1, q_2, p_k), \qquad u_k = \mathrm{or}(p_1, p_2, q_k), \qquad s_k = \mathrm{or}(p_0, p_1, q_k),$$

all nonzero by the nondegeneracy hypothesis.

Step 1. P_0 crosses Q_0 and Q_1, so the segments $[q_1, q_0]$ and $[q_1, q_2]$ share the endpoint q_1 and both cross $P_0 = [p_0, p_1]$. Applying Lemma 3.3 with $(A, B, C) = (q_0, q_1, q_2)$ and $[D, E] = P_0$ gives $v_1 w_1 < 0$.

Step 2. P_2 crosses Q_0 and Q_1, so those *same* two segments both cross $P_2 = [p_2, p_3]$. Lemma 3.3 at the same vertex q_1, now with $[D, E] = P_2$, gives $v_2 w_2 < 0$.

Step 3. P_1 crosses Q_0, so p_1 and p_2 lie strictly on opposite sides of the line $q_0 q_1$, i.e. $v_1 v_2 < 0$. Combining with Steps 1 and 2, w_1 and w_2 have opposite signs:

$$w_1 w_2 < 0. \tag{2}$$

Step 4. P_0 crosses all three of Q_0, Q_1, Q_2, which gives $s_0 s_1 < 0$, $s_1 s_2 < 0$ and $s_2 s_3 < 0$; in particular s_0 and s_2 have the same sign. Now $[p_1, p_0]$ and $[p_1, p_2]$ share p_1, and both cross Q_0 (because P_0 and P_1 each cross Q_0) and both cross Q_2 (because P_0 and P_1 each cross Q_2). Two applications of Lemma 3.3 at p_1 therefore give $s_0 u_0 < 0$ and $s_2 u_2 < 0$, whence u_0 and u_2 have the same sign. Since P_1 crosses Q_0 we have $u_0 u_1 < 0$, so

$$u_1 u_2 < 0. \tag{3}$$

Step 5. P_1 does *not* cross Q_1. By definition P_1 properly crosses Q_1 precisely when $u_1 u_2 < 0$ and $w_1 w_2 < 0$. By (3) the first holds, so the second must fail: $w_1 w_2 > 0$, contradicting (2). □

The proof uses each of the eight prescribed incidences and never the unconstrained cell (P_2, Q_2), which is what one would hope to see. Each of its inferences has additionally been checked as a universally quantified statement over all 3,315 order types of eight points and all 8! labellings, with no violation; see `independent/pi8_human/`.

For Π_{14} we still offer no human proof, and it is the weaker of the two pillars for exactly that reason: eleven points puts the order-type check of §3.2 permanently out of reach as well. A reader who wants to attack this work should attack Π_{14}.

For Π_8 we can also offer an independent verification by a completely different route, described next.

3.2 An independent check of Π_8 by order types

Π_8 constrains three consecutive sides of each polygon, so it involves exactly **eight points**, $P_0, \ldots, P_3$ and $Q_0, \ldots, Q_3$. Whether two segments cross is determined by orientation signs, hence by the *order type* of those eight points — and the order types of eight points in general

position are completely enumerated: there are 3315 of them [4, 3]. So Theorem 3.2 can be decided by enumeration, with no solver and no proof assistant.

We did this. Checking every labelling of every one of the 3315 representatives:

no order type of eight points admits Π_8 — 0 of 3315.

Two conventions might have left a gap and neither does. The database identifies mirror images, which is harmless here because reflection negates every orientation while each crossing test is a *product* of two, so the crossing relation is unchanged. And it stores one representative per order type, so the search enumerates labellings itself.

Remark 3.4 (Controls, and a by-product). "We found nothing" is worth nothing without a control that the search can find something. Two were run. The crossing pattern taken from the certificate of §6 *is* located in the database. Sharper: dropping any single one of Π_8's eight conditions yields a pattern that *is* realizable, and all eight relaxations are found. That rules out a vacuously failing search, and as a by-product it confirms Π_8 **is minimal** — a property we had previously established only by randomised minimisation of unsatisfiable cores, i.e. by the very machinery this check is meant to be independent of.

This is corroboration, not a second proof: it is independent of the Lean route in its source of truth (an externally published database), its algorithm (enumeration, not refutation), and its implementation.

Remark 3.5 (The same check cannot be run for Π_{14}). Π_{14} constrains four consecutive sides of P and five of Q, so it involves $5 + 6 = \mathbf{11}$ points, not eight. The enumeration of order types is complete only to $n = 10$ — and already 14,309,547 there — so this route is unavailable for Π_{14} and we do not have an independent exhaustive check of Theorem 3.7. Nor does the minimality of Π_{14} offer a way round: every proper subset of its conditions is realizable, so there is no smaller non-realizable sub-pattern to catch on fewer points.

3.3 The larger pattern

Definition 3.6. Pattern Π_{14} constrains $P_0, \ldots, P_3$ and $Q_0, \ldots, Q_4$ by fourteen incidences,

	Q_0	Q_1	Q_2	Q_3	Q_4
P_0	$*$	$\times$	$\times$	$*$	$\times$
P_1	$\times$	$\cdot$	$\times$	$\times$	$\times$
P_2	$\times$	$\times$	$*$	$*$	$\times$
P_3	$\times$	$\times$	$\times$	$*$	$*$

together with seven simplicity conditions: $P_0 \not\times P_2$, $P_0 \not\times P_3$, $P_1 \not\times P_3$, $Q_0 \not\times Q_2$, $Q_0 \not\times Q_3$, $Q_1 \not\times Q_3$, $Q_1 \not\times Q_4$.

Theorem 3.7. *No configuration realizes* Π_{14}.

Unlike Π_8, this one genuinely uses that the polygons do not cross themselves; the seven self-non-crossing conditions are part of the hypothesis and are supplied by simplicity.

3.4 How the patterns are proved non-realizable

Both theorems are established by encoding realizability as a Boolean satisfiability problem over the 364 orientation variables — Grassmann–Plücker relations, acyclicity, and the prescribed incidences — and refuting it. The refutation is not trusted:

1. Each clause of the encoding is *derived from the geometry* inside the proof assistant. The Grassmann–Plücker and acyclicity clauses are polynomial identities in the coordinates; the pattern clauses are the definition of crossing unfolded. Nothing about the encoding is assumed.

2. The solver's refutation is checked, then transcribed into an explicit chain of unit-propagation steps. The kernel verifies a chain of implications between propositions; *no search runs inside the proof assistant.*

Remark 3.8 (Controls against a vacuous encoding). The obvious worry is an over-constrained encoding, unsatisfiable for *everything*, which would refute the patterns for the wrong reason. Two things guard against it.

The substantive one: because every clause is *derived* from the geometry inside the proof assistant rather than asserted, a clause that did not in fact follow could not be proved at all. The general-position clauses are theorems about arbitrary configurations and they are satisfiable — the configuration of §6 has all 364 determinants nonzero, checked by exact arithmetic. So unsatisfiability cannot come from the general-position part; it must come from the prescribed incidences.

A cheaper check: the crossing matrix of the 38-crossing configuration of §6 contains *no* relabelled copy of either pattern. Note what has to be checked here — containment in Theorem 4.1 is up to the order-392 group, so observing that the identity labelling fails would establish nothing; all 588 instances must be excluded, and they are. This is a *negative* check that a vacuously unsatisfiable encoding would pass trivially; it is the first argument, not this one, that addresses that failure mode.

Remark 3.9 (Soundness, not completeness, is what is needed). The encoding is a set of *necessary* conditions for realizability. That is the right direction here: we are proving non-realizability, so we need only that every clause is valid. Completeness of the encoding — that every sign vector satisfying the clauses is realizable, which is false in general — is neither claimed nor used.

4 The covering theorem

Theorem 4.1. *Every crossing pattern of two simple heptagons with $I = 40$, and every one with $I = 42$, contains Π_8 or Π_{14} after relabelling.*

Two definitions are needed to read this statement.

A configuration *realizes* a pattern if its crossing relation agrees with every prescribed entry; entries marked $*$ are unconstrained. A crossing pattern *contains* Π if some relabelling carries Π onto a block of cyclically consecutive side indices of each polygon, on which all the prescribed entries hold.

"Relabelling" is the group of order $392 = 14 \cdot 14 \cdot 2$ generated by rotating and reflecting each polygon independently and exchanging P with Q (legitimate only because $m = n$). On *side* indices a vertex rotation by k sends P_i to P_{i+k}, whereas the reflection $p_i \mapsto p_{-i}$ sends P_i to P_{-i-1}; the shift matters when checking a containment by hand.

4.1 The $I = 42$ case

By (1), $I = 42$ forces $|E^D| = 7$, and with $\delta(G^D) \geq 1$ on both sides the graph is a perfect matching. There are $7! = 5040$ of these, falling into 28 orbits under the order-392 group. That every one of them contains Π_8 or Π_{14} is *machine-checked*, by a propagation chain of the same kind as Remark 4.2 describes. It is not a hand argument, and the brevity of this paragraph should not suggest otherwise. (The coincidence that 28 also counts the degree vectors of §4 below is exactly that — the two have nothing to do with each other.)

4.2 The $I = 40$ case, and its 28 degree shapes

Here $|E^D| = 9$, so G^D has nine edges with all fourteen degrees at least 1. Write $r_i = \deg P_i$, so $\sum_i r_i = 9$ with every $r_i \geq 1$. Subtracting one from each, the excess is 2 distributed over seven rows, so either

- one row has degree 3 and the rest degree 1 — 7 placements; or
- two rows have degree 2 and the rest degree 1 — $\binom{7}{2} = 21$ placements, which split by the cyclic distance 1, 2 or 3 between the two rows into $7 + 7 + 7$.

That is $7 + 21 = \mathbf{28}$ degree vectors, and up to rotation of P they are **four** shapes. The same applies to columns, and the column degrees enter through the constraint $\sum_j c_j = 9$ with $c_j \geq 1$.

Remark 4.2 (What is and is not human-sized here). The enumeration above is human-checkable and we have just done it. **The discharge of each of the 28 cases is not.** Each case is settled by a machine propagation chain in the same style as §3.4, drawing on a library of 588 pattern-avoidance lemmas together with the row- and column-degree facts; the four chains total 10,597 propagation steps. We state this explicitly because an earlier draft of this paper described each case as closed by "a short argument", and that was wrong.

Those 588 **lemmas are not additional inputs.** Each is a *relabelled copy* of Π_8 or Π_{14} under the order-392 group, and the count decomposes as $588 = 196 + 392$: the incidence matrix of Π_8 is symmetric under transposition, so the $P \leftrightarrow Q$ swap contributes nothing new for it, while Π_{14} gives the full 392. So "two patterns suffice" survives intact: the library is the two theorems of §3 instantiated, not new mathematics.

So Theorem 4.1 should be read as machine-checked, with the 28-fold case split being the part a reader can survey. What the reduction buys is not that the whole argument became hand-checkable, but that its *inputs* did: two pattern non-realizability facts, one of which is Definition 3.1.

5 Excluding 40 and 42

Corollary 5.1. *No configuration of two simple heptagons has $I = 40$ or $I = 42$.*

Proof. Immediate from Theorems 3.2, 3.7 and 4.1. □

Corollary 5.2. *Every configuration of two simple heptagons has $I \leq 38$.*

Proof. $I \leq 42$ by Lemma 2.3, I is even by Lemma 2.2, and 40 and 42 are excluded by Corollary 5.1. □

6 The lower bound

Proposition 6.1. *The heptagons with vertices*

$$P : (373, 157),\ (132, 5),\ (369, 182),\ (88, 157),\ (327, 282),\ (60, 272),\ (372, 298),$$
$$Q : (344, 24),\ (89, 303),\ (345, 19),\ (373, 214),\ (270, 338),\ (325, 93),\ (192, 284)$$

are simple, are in general position, and their boundaries meet in exactly 38 *points.*

All three assertions are decided by exact integer arithmetic and verified by the kernel. Combined with Corollary 5.2 this gives the equality in Theorem 1.2.

7 From orientation signs to the plane

Sections 3–5 are about sign vectors. Three further steps, together with the vertex–side reduction of §8.1, are needed to reach Theorem 1.2; all are proved and kernel-checked.

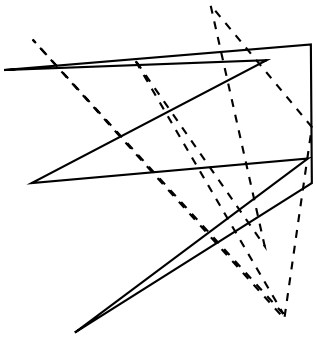


Figure 2: The configuration of Proposition 6.1: P solid, Q dashed. Their boundaries meet in 38 points. Drawn to scale from the integer coordinates.

7.1 Integer coordinates are no restriction

The combinatorial development uses integer coordinates. This costs nothing. Fix a configuration V with all 364 signs nonzero. Each sign condition is a strict inequality in the 28 coordinates, so the set of tuples with the same sign vector is open and contains a product of coordinate intervals around V, say of width $\delta > 0$. Since or is homogeneous of degree 2 that set is also scale-invariant, so tV is good for every $t > 0$ and the corresponding product of intervals around tV has width $t\delta$. Choose $t > 1/\delta$: every coordinate interval then has length exceeding 1 and so contains an integer, and rounding each coordinate independently lands on an integer point with the same 364 signs. No rationals and no clearing of denominators are required.

7.2 Counting points, not side pairs

The combinatorial argument counts crossing *pairs*; the theorem counts *points*. Under Hypothesis 8.1 these agree: a crossing pair meets in exactly one point, and conversely a shared point lying on a pair with a vanishing determinant would put a vertex on the other boundary, which (E9) forbids. The chain is then

$$\#\text{points}(V) \;=\; \#\text{pairs}(V) \;\le\; \#\text{pairs}(V') \;\le\; 38,$$

where V' is a perturbation as in §7.3. The equality and both inequalities are proved.

Theorem 1.2 does not assume (E9). Without it, a vertex of one polygon may lie in the interior of a side of the other, and a point of $\partial P \cap \partial Q$ can then lie on no crossing pair. That case is handled by the vertex–side reduction of §8.1, which bounds the number of points by the number of crossing pairs plus vertex–side incidences and removes the incidences one at a time without decreasing that sum, ending at a configuration with none, where the chain above applies.

7.3 Removing the collinearity assumption

[2, §3] assumes, "for convenience" and without proof, that no three of the combined vertices are collinear. We do not assume it. The argument is a perturbation, in two halves:

Lemma 7.1. *Crossing is an open condition, so for a small enough perturbation of the fourteen vertices no existing crossing is lost.*

Lemma 7.2. *If two segments are disjoint, they remain disjoint when their four endpoints move by less than some $\varepsilon > 0$.*

Lemma 7.1 handles the crossings; Lemma 7.2 handles simplicity. These need *different* arguments: for a *given* pair of sides, "does not cross" is a closed condition, so it cannot be inherited

from the openness of crossing that gives Lemma 7.1. Simplicity is nonetheless stable, because for a polygon simple as a point set the non-adjacent closed sides are disjoint compact sets and hence at positive distance — which is what Lemma 7.2 exploits. The configurations in general position are dense: the complement is the zero set of finitely many nonzero polynomials in the 28 coordinates, hence a finite union of hypersurfaces. Combining this with the two lemmas approximates an arbitrary pair of point-set simple heptagons by one in general position with at least as many crossings, giving the bound for the original. A single ε must serve every crossing, every disjoint side pair and all 364 determinants at once; since there are finitely many conditions, one takes the minimum.

One case deserves separate mention, since Lemma 7.2 does not cover it: *adjacent* sides share an endpoint and so are never disjoint. Simplicity constrains them too — they must meet only at that shared vertex — and this is exactly the delicate case, because three consecutive collinear vertices are what §7.3 exists to permit. In the formal development the adjacent case is handled separately, by the observation that once the perturbed configuration is in general position its adjacent sides cannot overlap. Lemma 7.2 itself is proved for an arbitrary real normed space and is independent of everything else here.

Remark 7.3. Note the direction: we need the perturbed configuration to have *at least* as many crossings, not the same number, and openness gives exactly that.

8 The vertex-coincidence reduction

Theorem 1.2 uses no external result. This section concerns only the further step from Theorem 1.2 to $f(7,7) = 38$. The conclusion of [2, §9] is the following, and earlier versions of this paper assumed it whole.

Hypothesis 8.1 ((E9))**.** No vertex of either polygon lies on the boundary of the other.

Its proof splits in two: §9.1 removes vertex–vertex coincidences and §9.2 removes vertex–side ones. The second half is proved in §8.1, so Theorem 1.2 needs only the property corresponding to the first:

Hypothesis 8.2 ((E9.1), no shared vertex)**.** No vertex of P is a vertex of Q.

Remark 8.3 (A hypothesis and a reduction are different things). Theorem 1.2 is unconditional under its hypotheses, and Hypothesis 8.2 is one of them. The deduction of $f(7,7) = 38$ needs something else: a *reduction*, stating that every pair can be replaced by one satisfying Hypothesis 8.2 and the other hypotheses of Theorem 1.2 without losing intersection points. We state it with explicit quantifiers, since this is the precise statement we use.

Proposition 8.4 (External; [2, §9.1])**.** *For every pair (P, Q) of simple heptagons with finitely many boundary intersections there is a pair (P', Q') of simple heptagons, again with finitely many boundary intersections, satisfying Hypothesis 8.2 with $|\partial P' \cap \partial Q'| \geq |\partial P \cap \partial Q|$.*

Remark 8.5 (Only §9.1 is needed). Proposition 8.4 asks only for vertex–vertex coincidences to be removed. It does *not* ask that the perturbed pair avoid vertices lying in the interiors of sides, which is what §9.2 would supply and what Theorem 1.2 does not need. A reader comparing this paper with [2] should compare it with §9.1.

Corollary 8.6. *Assume Proposition 8.4. Then $f(7,7) = 38$.*

Proof. Given any pair of simple heptagons whose boundaries meet in finitely many points — which is the range over which $f(7,7)$ is defined — Proposition 8.4 produces one satisfying Hypothesis 8.2, still with finitely many boundary intersections and with at least as many intersection points, and Theorem 1.2 bounds that by 38; the bound is attained by Proposition 6.1. □

Proposition 8.4 is the only external mathematical input to Corollary 8.6, and we do not prove it. The following remarks record its source and the extent to which we have checked its applicability.

Remark 8.7 (Provenance). We read [2] in its journal version, which is the only version containing §9: the SoCG/arXiv version [1] ends at §8 and contains no treatment of degeneracies at all. A referee checking our citation against arXiv will not find the section.

Remark 8.8 (What §9 says, in its own words). Since we ask the reader to accept that Hypothesis 8.1 is what §9 delivers, we quote it. [2, §9] opens:

> "We will now justify our general-position assumption that we introduced in Sect. 3 stating that *no vertex of one polygon lies on the boundary of the other polygon*. That is, we will show that one can get rid of such degenerate cases *without decreasing the number of intersection points*."

The second clause is the inequality of Proposition 8.4. The other conditions of the Proposition — that P' and Q' are again simple and again meet in finitely many points — are not in that sentence; see Remark 8.10. Their §3 makes finiteness a standing assumption ("the basic assumption that is made throughout the paper is that P and Q have only finitely many intersection points"), and their notion of a simple polygon is that the sides are pairwise disjoint except for the common endpoints of consecutive sides, which corresponds to (S2)–(S3) of Appendix A, with (S1) implicit.

Remark 8.9 (Their standing assumption is broader, and why that is harmless). [2] assume throughout that no three of the combined vertices are collinear *and* that no two of the combined sides are parallel. We use neither. One might worry that §9's own argument runs under that full assumption, in which case Proposition 8.4 would be stronger than what §9 delivers, and §7.3 would buy nothing at the level of the final claim.

It does not, and their §3 is explicit about the order. It first states the vertex condition, "justified in Sect. 9 by a non-trivial perturbation"; and only then, *given* that condition,

> "we assume *for convenience* that no three vertices of P and Q combined lie on a line, and no two sides of P and Q combined are parallel. Indeed, given that no vertex of one polygon lies on the other polygon, this can be achieved easily by another small perturbation of the sides without changing the intersection pattern."

So §9 is *prior* to the collinearity and non-parallelism clauses rather than dependent on them, and §7.3 does remove the former from the final claim. The latter we never use.

The quotation also shows that this second perturbation is stated without a detailed argument: "for convenience", "can be achieved easily", with no lemma number. That is the step §7.3 proves, and it is why we treat it separately from §9 rather than grouping both under "general position".

Remark 8.10 (Applicability: what this project has not settled). Theorem 1.2 assumes that the polygons are simple as point sets, that their boundaries meet in finitely many points, and Hypothesis 8.2, so the reduction must deliver all three, together with the count inequality. §9.1 states the count inequality. We have not found an explicit statement in [2] that the perturbed polygons remain simple, or that the perturbed boundaries again meet in finitely many points. Both are natural to expect of a sufficiently small perturbation, and this project has partial results: a kernel-checked lemma that simplicity, in the sense of Appendix A, survives every sufficiently small linear motion of the vertices (`SimplicityStable`); and unreviewed pen-and-paper analysis of how the §9.1 directions can be chosen so that the intersection stays finite. These have not been assembled with §9.1 into a proof of Proposition 8.4 for the definitions used here. Until that is done, or confirmed by a reader of [2], we state the applicability of §9.1 to Proposition 8.4 as unresolved in this project. We do not suggest that the result of §9.1 is in doubt.

Why the reduction is needed, precisely: without it, a point of $\partial P \cap \partial Q$ can be a shared vertex, and the counting argument of §8.1 does not apply there — a shared vertex is one point but four vertex–side incidences, so the measure that drives §8.1 does not decrease when the coincidence is removed.

Remark 8.11 (Citation). We do not attribute all of the usual general-position package to §9. §9 gives Hypothesis 8.1. The collinearity clause rests on one sentence of §3 and is proved here (§7.3). Simplicity and finiteness of the perturbed pair are discussed in Remark 8.10.

8.1 The vertex–side half is proved

Hypothesis 8.1 decomposes into a vertex–vertex half (§9.1) and a vertex–side half (§9.2). **The vertex–side half is proved here**, in Lean, by an argument of our own (a "tilted" shift) rather than by formalizing §9.2 as printed. The vertex–vertex half is not proved; it is Hypothesis 8.2 of Theorem 1.2, and the corresponding reduction is Proposition 8.4.

What is proved. If a vertex of Q lies in the *interior* of a side of P, it can be removed by a shift, and iterating terminates. The primitive the argument needed is elementary: the signed area against a fixed line is *affine* along a segment, so a nonzero affine function of one real variable has at most one root, and hence a segment not contained in a line meets that line at most once. From this, on a side touched by the shift, every determinant against every vertex of Q is nonzero afterwards — those already nonzero stay nonzero, and those that vanished become δ times a nonzero rate — so that side meets each side of Q at most once; an untouched side has not moved, and neither has Q, so its meetings are exactly the original ones. The intersection is a finite union of the 49 side-pair meetings, so it stays finite. The decreasing step therefore reproduces *both* of its own hypotheses, and the strong induction on the number of vertex–side incidences runs to a degeneracy-free endpoint. This is what upgrades Theorem 1.2 from (E9) to (E9.1).

What is not, and two negative observations. The vertex–vertex half resists, and the obvious routes are closed off:

- Generic perturbation preserving proper crossings does *not* imply that the point count is monotone.
- Naive displacement into the interior of a cone does not either.

Explicit coordinates exhibiting both failures are recorded in the project repository; they are analytic and *not* machine-checked. This is consistent with the case on which an earlier attempt of ours broke: writing $\mathrm{Cone}_P(v)$ for the cone spanned at a vertex v by its two incident sides, the difficulty arises when $\mathrm{Cone}_P(v) \subseteq \mathrm{Cone}_Q(w)$, since then displacing v into the cone at w can destroy the intersection point rather than split it.

There is a further, sharper obstruction, and it is specific to the measure that made §9.2 work. That induction decreases the number of vertex–side incidences; but a single vertex–vertex coincidence is *one* point while being *four* incidences, so the same measure does not decrease across a coincidence — on a committed witness configuration, every admissible move drops it by exactly 3. The natural repair is to count intersection *points* rather than incidences, and the exact identity relating the two is proved; but that the point count is monotone along the required moves is supported only numerically and is *not* proved. None of this is evidence that AKR's theorem is false, and we allege no error in it; what remains is substantial unfinished work.

9 Machine verification, and its limits

The mathematical assertions of §§2–7, the positive results of §8.1, and Theorem 1.2 are verified by the Lean 4 kernel. The build log of the headline target (§9.2) records no `sorry`; the axiom check finds no `native_decide` (`Lean.ofReduceBool`) and no project-specific axiom among the dependencies of the declarations it inspects, each of which depends on at most the three standard axioms `propext`, `Classical.choice`, `Quot.sound`. Hypothesis 8.2 is a hypothesis of the statement, never an axiom. Kernel checking establishes the formal statements under their stated hypotheses and these foundational assumptions, relative to the correctness of Lean and of the Mathlib definitions used (for example `segment` and `Set.ncard`). It does not establish that the formal statements mean what this paper says they mean; that is Remark 9.1.

Remark 9.1 (What the kernel does not check). The kernel proves theorems about definitions; it does not certify that the definitions mean what we say. That $\partial P \cap \partial Q$ is modelled by the right set, and that "simple heptagon" is modelled by the right predicate, are judgements a human reader must make. This is, in our view, the principal remaining risk, and it is not reducible by more computation.

One partial check is available and we have carried it out: the definitions, applied to the explicit 38-crossing configuration of Proposition 6.1, yield exactly 38. A wrong definition would have to be wrong in a way that still lands on that value.

One modelling choice deserves mention. A heptagon is formally a cyclic sequence of seven vertices satisfying (S1)–(S3); three consecutive vertices may be collinear, so a "heptagon" with a straight vertex is admitted. This enlarges the class over which the upper bound is proved, and the lower-bound example has no straight vertex, so it does not affect the value 38.

9.1 Which results a human can check, and which are machine-only

We think a referee is owed this table explicitly rather than having to infer it.

Result	**Status**	**Checkable by hand?**
$I + \lvert E^D \rvert = 49$, Lemma 2.2, Lemma 2.3	machine-checked	**yes**, and short
The 28 degree shapes (§4)	machine-checked	**yes**; derived above
Theorem 3.2 (Π_8 non-realizable)	machine-checked	**yes** — proof via Lem. 3.3
Theorem 3.7 (Π_{14})	machine-checked	**no** — solver refutation
Theorem 4.1 (covering)	machine-checked	**no** — see Rem. 4.2
Proposition 6.1 (the 38 example)	machine-checked	**yes** in principle; 364 determinants
§7 (reduction to the plane)	machine-checked	**yes**; ordinary analysis
§8.1 (vertex–side half)	machine-checked	long, but elementary steps
Theorem 1.2	machine-checked	by assembling the rows above
Proposition 8.4	**external, not formalized**	[2, §9.1]; see Rem. 8.10

Hypothesis 8.2 does not appear as a row: it is a hypothesis of Theorem 1.2, not a claim. Proposition 8.4 is used only for Corollary 8.6.

The two "no" rows are the substance of the exclusion. What the reduction achieves is that the machine-only part now rests on *two* non-realizability facts about small explicit patterns instead of on an enumeration of 13,003,620 objects — and one of those two, Π_8, is small enough to state in a 3×3 table (Figure 1).

9.2 How to reproduce

The development is in the repository
`https://github.com/NewYorkImperialist/heptagon-intersections`,

which at the date of this revision is private (§12). Its Lean sources have not changed since commit `a211140` (24 September 2026). The toolchain is pinned to Lean 4.33.1 in `lean/lean-toolchain`, and Mathlib to tag `v4.33.1` (revision `0df444a`) in `lean/lake-manifest.json`. The following builds the headline theorems and checks their axiom dependencies:

working directory	command
`lean/`	`lake exe cache get`
`lean/`	`lake build Headline 2>&1 | tee ../headline-build.log`
repository root	`bash scripts/verify-all.sh --axiom-log headline-build.log`

The first command downloads prebuilt Mathlib files for the pinned revision. The second builds the target `Headline`: 332 modules of this project (the 331-module import closure of `MainTheorem`, `RealUpperBound`, `AKR9` and `Pair9Shift`, plus `Headline` itself), reported by Lake as 3,344 jobs because Mathlib jobs are counted too. The third parses the `#print axioms` output in the log, needs no Lean, and exits 0 only if no declaration depends on an axiom other than the three above, `sorryAx` appears nowhere, and each of six named headline theorems printed its own axiom line.

A bare `lake build` is *not* sufficient: the package's default targets omit `AKR9`, `Pair9Shift` and `Headline`. `bash scripts/verify-all.sh` with no arguments builds both the default targets and `Headline` and runs the repository's other checks; a cold run of it is dominated by the 1,084 `OTEnum*` and `Strat*` enumeration modules of an earlier approach that the headline theorems do not import, and takes many CPU-hours.

The results of this paper correspond to the following declarations (all in the namespace `Heptagon`).

In this paper	In the development
Theorem 3.2	`MinPatternClauses000.minpattern000_impossible`
Theorem 3.7	`MinPatternClauses001.minpattern001_impossible`
Theorem 4.1 with the above, $I = 40$	`Cover.no_nine_edges`
the same, $I = 42$	`Cover.no_seven_edges`
Corollary 5.1	`ConfigBridge.crossCountC_ne_40`, `_ne_42`
Lemma 7.2	`SegmentStable.exists_eps_disjoint_stable`
pair count, no general position	`RawConfig.raw_crossPairs_card_le_38`
Theorem 1.2	`Pair9Shift.points9_le_38_of_coincFree`
with attainment	`Pair9Shift.isGreatest_38_of_coincFree`
Hypothesis 8.2	the `hcoinc` argument of the above
general-position case	`RealUpperBound.isGreatest_38_real`
Hypothesis 8.1	`AKR9.NoVertexOnOther`
(E9) version, superseded	`AKR9.points_le_38_of_no_vertex_on_boundary`
and its equality	`AKR9.isGreatest_38_of_E9`

The pattern and covering theorems are stated for integer-coordinate configurations with all 364 determinants nonzero; §7 transfers them to the plane. `SegmentStable` imports only Mathlib, so Lemma 7.2 can be read on its own.

Recorded builds. The repository's GitHub Actions workflow runs the three commands above (with the Mathlib download but no cache of the project's own build) on a GitHub-hosted runner, together with eight tests confirming that the axiom checker rejects deliberately corrupted logs. It passed on commit `b78f372` (run `36086080499`, 25 September 2026) and on commit `e984888` (run `36089101301`, the same day); both commits have the Lean sources of `a211140`. These runs are records of that workflow at those revisions. They are not independent mathematical review, and they do not cover later changes to the repository.

The solver refutations are shipped as certificates and re-checked by a standalone proof checker independent of the proof assistant; that checker is itself first shown to reject non-refutations.

9.3 Independent corroboration

Beyond the kernel: the terminal case analysis of an earlier version of this argument was reproduced by a second, independently written implementation in a different language; all solver refutations are re-checked by a standalone proof checker, which is itself first shown to reject non-refutations; and all generated artifacts are regenerated and compared byte-for-byte. These guard against implementation error. *None of them is a check on whether the argument is the right argument.*

10 Related work

Dillencourt, Mount and Saalfeld [6] posed the problem and settled the even cases. Černý, Kára, Král', Podbrdský, Sotáková and Šámal [5] gave bounds for odd cases and solved $k = 5$, explicitly leaving $k = \ell = 7$ open. Ackerman, Keszegh and Rote [2] obtained asymptotically optimal bounds; their §9.1 is the source of Proposition 8.4. Günther's preprint [7] on odd n was withdrawn by its author.

Both of our tools are standard. Encoding order-type or chirotope realizability as a satisfiability problem and refuting it mechanically is well established, including in combination with proof assistants. Framing extremal results in terms of forbidden configurations is likewise standard. What we have not found in the literature is either applied to crossing patterns of two simple polygons, or to $f(m, n)$.

Remark 10.1 (On novelty). We searched for prior statements of the covering result and found none. We record plainly that this is **weak evidence and not a novelty claim**: the search was bounded, the forward citations of [5] were not examined, and the relevant literature is small enough that a result could sit in it under different vocabulary. A domain expert's judgement is not replaced by such a search.

11 Further work: the family $f(7, n)$

We record an observation which suggests that the natural sequel to this work is not $f(9, 9)$ but the whole family $f(7, n)$, and which costs two lines.

G^D has $m + n$ vertices and, by Lemma 2.3, no isolated ones. A graph on N vertices with c connected components has at least $N - c$ edges, so $|E^D| \geq (m + n) - c$ and, with (1),

$$I \;\leq\; mn - (m + n) + c.$$

For m, n both odd, $mn - (m + n)$ is odd while I is even. Hence $c \leq 4$ already forces $I \leq mn - (m + n) + 3$, the conjectured value: **to prove the conjecture for a given** (m, n) **it suffices to bound the number of components of** G^D **by** 4**.**

Now fix $m = 7$. Every component of G^D contains an edge and therefore a side of P, and there are only seven of those, so $c \leq 7$ *automatically, for every* n. Consequently

$$f(7, n) = 6n - 4 \text{ for all odd } n \quad \text{would follow from ruling out } c \in \{5, 6, 7\},$$

three cases whose number does not grow with n. Theorems 3.2, 3.7 and 4.1 of this paper do exactly that for $n = 7$: $I = 42$ forces $c \geq 7$ and $I = 40$ forces $c \geq 5$.

This places the problem on a visible ladder. At $m = 5$ only $c = 5$ has to be excluded — a single case — and that is precisely the range [5] settle, obtaining $f(5, \ell) = 4\ell - 2$ for every ℓ. So

$m = 7$ is the next rung above a solved problem, at three cases rather than one, and this paper supplies its base case.

Two further remarks, one encouraging and one not.

Remark 11.1 (The patterns are not about heptagons). Neither Definition 3.1 nor Definition 3.6 mentions the number of sides: Π_8 is a statement about eight points and three consecutive sides of each polygon, and Π_{14} about eleven. Theorems 3.2 and 3.7 are therefore already theorems about n-gons for every n, and would be reused unchanged. What does not transfer is Theorem 4.1, the covering claim, which is verified by enumeration over a domain that depends on n.

Remark 11.2 (Why a single pattern cannot suffice, for any fixed n-free argument). The case $c = 7$ is rigid: each component holds exactly one side of P, so every side of Q fails to cross *exactly one* side of P, giving a surjection σ from the sides of Q onto those of P, where $\sigma(j)$ is the unique side of P that Q_j misses. Consider the diagonal $\sigma(j) \equiv j \pmod 7$, with j taken in $\{0, \ldots, n-1\}$.

Suppose Π_8 occurred on P_i, P_{i+1}, P_{i+2} and Q_j, Q_{j+1}, Q_{j+2}. The prescribed non-crossing at the centre gives $\sigma(j+1) = i+1$. If $\sigma(j+1) = \sigma(j) + 1$ in $\mathbb{Z}/7$, this forces $i = \sigma(j)$, so (P_i, Q_j) is a non-crossing as well — contradicting the crossing that Π_8 prescribes in that corner. **That last step is where an earlier version of this remark overreached:** $\sigma(j+1) = \sigma(j) + 1$ **holds for every window except the one in which** j **wraps from** $n-1$ **to** 0**, and at that seam it holds precisely when** $7 \mid n$**.** So the argument kills every window except the one at $j = n-1$ unconditionally, and that one only under a hypothesis on n.

The seam window is small enough to settle outright. There $j = n-1$ and $\sigma(0) = 0$, so the centre condition forces $i = 6$; the remaining prescribed crossings reduce to the single requirement $\sigma(n-1) \notin \{6, 0, 1\}$: the remaining cells ask $\sigma(0) \notin \{6, 1\}$ and $\sigma(1) \notin \{6, 0\}$, and $\sigma(0) = 0$, $\sigma(1) = 1$ satisfy those automatically, while the corner (P_1, Q_1) is the unconstrained $*$ of Definition 3.1. Hence

$$\text{the diagonal avoids } \Pi_8 \iff n \equiv 0, 1, 2 \pmod 7.$$

In particular the obstruction does hold whenever $7 \mid n$, and it happens to hold at $n = 9$; but it **fails** at $n = 11$ and at $n = 13$, where the diagonal does contain Π_8 — in a single window read in the increasing direction, the one straddling the seam, at $(i, j) = (6, 10)$ and $(6, 12)$ respectively. (Reading windows in the decreasing direction instead yields one further copy at each of those n, also across the seam, and none at all when $n \equiv 0, 1, 2$.) We confirmed the equivalence by direct enumeration for every $n \leq 120$ and both reading directions, with two independent implementations and with controls confirming the search is not vacuous. The experiment recorded in `independent/pattern_scaling/` tests exactly this claim and reaches the same conclusion; it adds that the $n = 9$ diagonal, which Π_8 does miss, is caught by Π_{14}.

The conclusion of the remark survives this repair, but on a restricted family rather than at every n: the diagonal avoids Π_8 for every $n \equiv 0, 1, 2 \pmod 7$, which is infinitely many n and infinitely many odd n, so no argument for all n can rest on Π_8 alone. It would need a family of patterns closed under this kind of structure, or a different idea altogether. We do not know whether a *finite* family suffices for all n, and if the number required grows with n then this route fails.

We make no claim that any of this is easy. The point is only that the obstruction is identifiable and the target is a single, finitely stated combinatorial question, and that a general argument — not an enumeration — would be needed, since "all n" is not something a computation reaches.

12 Status, and what we ask of a reader

To state the position exactly:

- **Kernel-checked:** Theorem 1.2 — for simple heptagons sharing no vertex whose boundaries meet in finitely many points, at most 38 boundary intersection points, and 38 attained — together with every assertion of §§2–7 and the positive results of §8.1, with no published result about $f(m, n)$ used as an input. This includes all pairs in general position.
- **Using an external theorem:** $f(7, 7) = 38$ follows from Theorem 1.2 and Proposition 8.4, which is [2, §9.1] and is not formalized here.
- **Unresolved in this project:** whether §9.1, for the definitions of Appendix A, also yields a pair that is again simple and again has finite intersection (Remark 8.10).
- **Review:** no independent review of the mathematics or of the formal development has been completed. The project's records note that a colleague began reading an earlier dependency manifest in September 2026; no findings from that reading are recorded.
- **Access:** as of 4 October 2026 the repository, and with it the recorded GitHub Actions runs, is private; access is by invitation.

12.1 What can be judged without a proof assistant

The machine-checked proofs in this work are long generated propagation chains, and they are not intended to be read line by line. A reader without Lean experience can nonetheless assess several parts that call for a mathematician's judgement: the statements and definitions, set out in ordinary notation in Appendix A; the human-readable arguments of §§2–7; and the external reduction. Checking that Appendix A faithfully transcribes the Lean definitions, that the formal statements are the ones described, and that the recorded builds refer to the code described, is easier with Lean experience. The accompanying audit document (`docs/audit/audit.pdf`) quotes the formal statements verbatim beside their mathematical reading to support that check, but it does not replace review by a reader of Lean.

Three questions are the ones we would most value:

1. **Is Theorem 3.7 true?** Twenty-one incidences on eleven points (Definition 3.6). This is a question of plane geometry, and it is now the *only* one of the two patterns without a human proof: Theorem 3.2 acquired one via Lemma 3.3, but eleven points puts both that argument and the order-type check of §3.2 out of reach here. A synthetic argument — or a counterexample — would be decisive either way. If it is false, the exclusion of 40 and 42 collapses, though §2, §6 and §7 would survive.
2. **Does [2, §9.1] deliver Proposition 8.4 for the definitions used here?** This is the one external input to Corollary 8.6, and settling it is a matter of reading that paper together with Appendix A. One point we were able to check ourselves: §9 does *not* depend on AKR's standing collinearity and non-parallelism assumptions, because their §3 introduces §9's vertex condition *before* those clauses and derives them from it (see the remark on their standing assumption in §8). What remains is recorded in Remark 8.10: §9.1 states the intersection-count inequality, and we have not found an explicit statement there that the perturbed polygons remain simple and meet in finitely many points.
3. **Is the covering result (Theorem 4.1) already known?** Our search found nothing, and we make no novelty claim (Remark 10.1).

We would also welcome an opinion on Appendix A — whether those definitions are the intended objects — which is the risk Remark 9.1 identifies and which further computation cannot settle.

Disclosure of generative-AI use

Generative-AI systems were used substantially throughout: exploring the reduction, generating essentially all software including the formalization, auditing the cited sources, and drafting this paper. The named author is responsible for the claims. AI output was not treated as trustworthy: each major claim is checked by a separately recorded mechanism — source audit, independent reimplementation, a machine-checkable certificate with a standalone checker, or kernel verification — and errors introduced during development are catalogued in the project repository rather than quietly repaired. The patterns of §3 were *found* by computer search; what they are is small enough to check by hand.

A The formal definitions, in ordinary notation

Remark 9.1 identifies definitional faithfulness as the principal residual risk: the kernel proves theorems *about* these definitions and cannot certify that they are the intended objects. That judgement is a reader's, so we set them out here. These are transcriptions of the formal statements, not paraphrases; indices are mod 7 and $[x, y]$ denotes the closed segment.

Let $P, Q : \mathbb{Z}/7 \to \mathbb{R}^2$.

Simple as a point set. P is *simple* if
(S1) $P_i \neq P_{i+1}$ for every i (no degenerate side);
(S2) $[P_i, P_{i+1}] \cap [P_j, P_{j+1}] = \emptyset$ whenever P_i and P_j are non-adjacent sides;
(S3) $[P_i, P_{i+1}] \cap [P_{i+1}, P_{i+2}] = \{P_{i+1}\}$ for every i.
Nothing here constrains collinearity: three consecutive vertices of P may be collinear, and so may any three of the fourteen.

What is counted. The boundaries and their intersection are

$$\partial P = \bigcup_i [P_i, P_{i+1}], \qquad \partial Q = \bigcup_j [Q_j, Q_{j+1}], \qquad X = \partial P \cap \partial Q,$$

genuine subsets of $\mathbb{R}^2$, and the quantity bounded is the cardinality of X.

The hypotheses of Theorem 1.2. Besides (S1)–(S3) for P and for Q:
(H1) X is finite;
(H2) $P_i \neq Q_j$ for all i, j (no shared vertex, Hypothesis 8.2).
(H1) is an explicit hypothesis, and it matters: the formal cardinality function returns 0 on infinite sets, so without (H1) a bound would say nothing about overlapping boundaries.

The theorem. With P, Q satisfying (S1)–(S3), (H1) and (H2), $|X| \leq 38$; and 38 is attained, by Proposition 6.1.

The older hypothesis (E9). Hypothesis 8.1, used by the superseded (E9) version, is

$$P_i \notin [Q_j, Q_{j+1}] \quad \text{and} \quad Q_j \notin [P_i, P_{i+1}] \qquad \text{for all } i, j.$$

It implies (H2), and under it (H1) is a consequence rather than a hypothesis: X then injects into the set of crossing side pairs.

The question we are asking is whether (S1)–(S3), (H1), (H2) and X above are the objects a geometer means by "two simple heptagons sharing no vertex whose boundaries meet in k points". If they are not, machine checking does not help.

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